

1.7 Human-Computer Interaction Problems

ایچ سی آئی مسائل وہ رکاوٹیں ہیں جو سسٹم کو انسانی ضرورت یا ذہنی حد کے مطابق نہ ہونے پر پیدا ہوتی ہیں۔

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 10 of 16

★ Golden

1.7 Human-Computer Interaction Problems

★ Golden

LEARN IT

HCI problems are common difficulties that stop users from using technology smoothly. They happen when systems fail to meet human needs or cognitive limits. **Cognitive limits** means the mind can only hold and process so much at once — if an app asks you to remember three SMS codes at the same time, it has gone past that limit.

Board answers score well when you name the problem, explain it in one line, and give a concrete example (the source examples are safe; a matching local example is even better).

Problem	Explanation and example
Interface complexity	Too many options or a confusing layout. Example: a shopping app with so many menus you cannot find the checkout button.
Poor feedback	The system does not show if an action worked or failed. Example: clicking Submit and the screen freezes with no confirmation.
Inconsistency	Different parts of the app look or act differently. Example: a magnifying-glass icon for Search on one page, a binocular icon on another.
Accessibility gaps	Design ignores people with disabilities. Example: a video tutorial platform without captions for deaf users.
Compatibility issues	Interface fails on different devices or browsers. Example: a website that works on a laptop but crashes on a mobile browser.
Contextual neglect	The system does not adjust to the user's situation. Example: a map app with no dark mode, blinding the driver at night.
High cognitive load	Users must memorize too much or follow overly complex steps. Example: a registration system forcing you to remember three different SMS codes at once.
Trust deficiency	The system feels unsafe or hides security cues. Example: an online store asking for credit cards but lacking an SSL/HTTPS lock icon.

WORKED EXAMPLE

Worked example: diagnose three local systems.

ATM: six tiny unmarked buttons beside the screen and no "Cash is coming" message → interface complexity + poor feedback. A user walks away while cash is still coming out.

JazzCash: "Send" is green on the home screen but orange on the confirm screen, and there is no padlock on the payment page → inconsistency + trust deficiency.

School portal: works in Chrome on a lab PC, blank on a student's Android browser; exam instructions are a caption-free video; login needs a portal password, an email OTP and an SMS OTP on the same screen → compatibility, accessibility gap, high cognitive load.

Once you can name the problem, Section 1.8 tells you how to fix it.

REMEMBER IT

Eight problems in two families: (1) Screen problems — complexity, inconsistency, accessibility, compatibility, context. (2) Mind-and-trust problems — poor feedback, cognitive load, trust. If the user is lost, look at the screen family. If the user is scared or exhausted, look at the mind family.

COMMON MISTAKE

Mistake: Writing "the app is bad" as an HCI problem, or mixing up cognitive load with compatibility.

Fix: Use the exact names from the table. Cognitive load is "too much to remember". Compatibility is "breaks on another device or browser".

اردو میں سمجھیں

ایچ سی آئی مسائل وہ رکاوٹیں ہیں جو سسٹم کو انسانی ضرورت یا ذہنی حد کے مطابق نہ ہونے پر پیدا ہوتی ہیں۔ آٹھ مسائل: انٹرفیس کی پیچیدگی، کمزور فیڈبیک، عدم مطابقت (ایک جیسا نہ ہونا)، رسائی کی کمی، آلہ/براؤزر سے ناسازگاری، سیاق کو نظرانداز کرنا، زیادہ ذہنی بوجھ، اور اعتماد کی کمی (جیسے HTTPS تالے کے بغیر کارڈ مانگنا)۔

EXAM TIP

Common 8-mark question: "Describe problems of HCI with examples." Aim for at least six of the eight, each in the pattern Name → one-line meaning → example. Memorise the source examples (checkout buried in menus, frozen Submit, binocular vs magnifying glass, no captions, laptop-only site, no dark mode, three SMS codes, missing HTTPS lock). Adding ATM / JazzCash / portal examples shows application and often gains the last marks.

CHECK YOURSELF

1. Which problem is a missing HTTPS lock on a payment page?
2. Why are three SMS codes at once an HCI problem?
3. A site works on a laptop but crashes on a phone. Name the problem.

Answers: 1. Trust deficiency. 2. High cognitive load — users must memorize too much. 3. Compatibility issues.

Next lecture: 1.8 Methods for Improving HCI